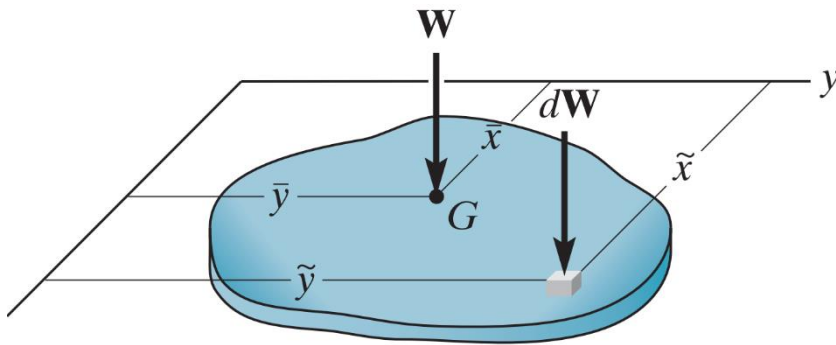
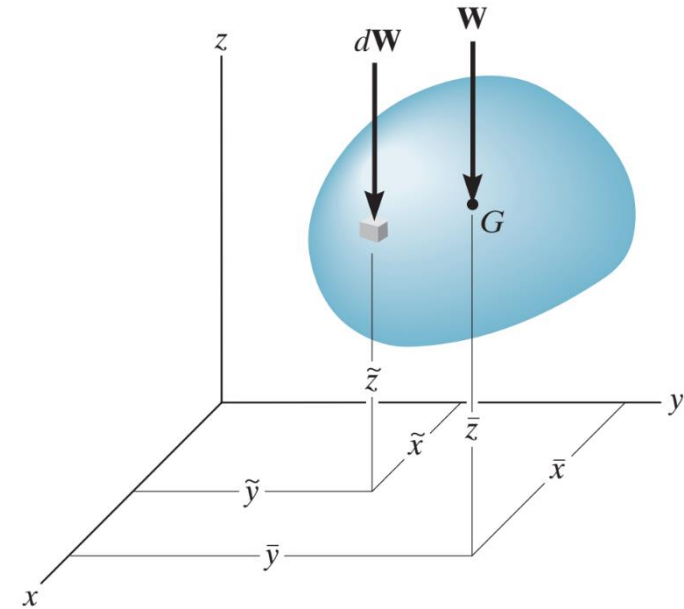
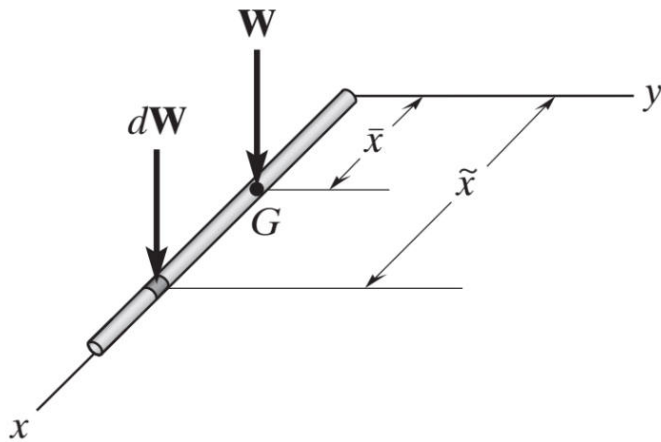


Gebze Technical University
Dep. of Mechanical Engineering

Statics (ME 231)

Lecture 15:
Center of Gravity and Centroids
By Saeed Lotfan

Center of Gravity, G



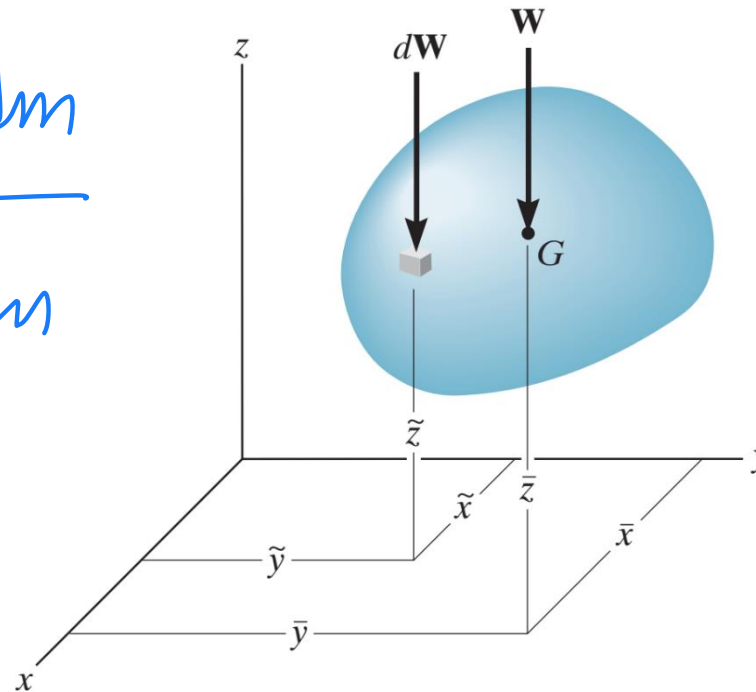
$$W = \int dW$$

Center of Gravity, G

$$\bar{x} = \frac{\int \tilde{x} dW}{\int dW} \quad \begin{array}{l} \text{pink arrow} \rightarrow dW = g dm \\ \text{blue arrow} \rightarrow \int \tilde{x} g dm \\ \text{blue underline} \rightarrow \int g dm \end{array}$$

$$\bar{y} = \frac{\int \tilde{y} dW}{\int dW}$$

$$\bar{z} = \frac{\int \tilde{z} dW}{\int dW}$$



Center of Mass, C_m

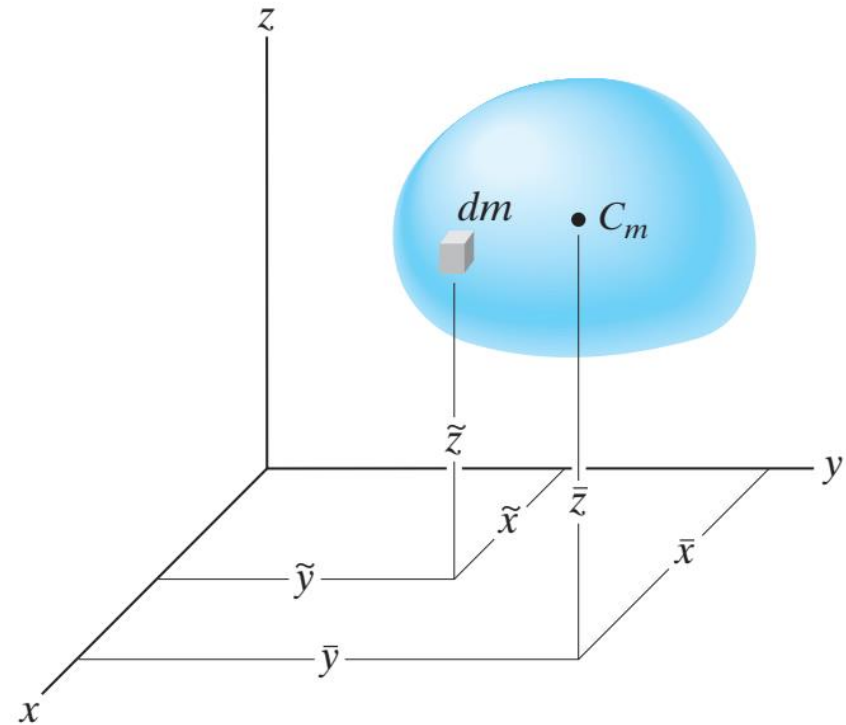
$$\bar{x} = \frac{\int \tilde{x} \, dm}{\int dm}$$

$$\bar{y} = \frac{\int \tilde{y} \, dm}{\int dm}$$

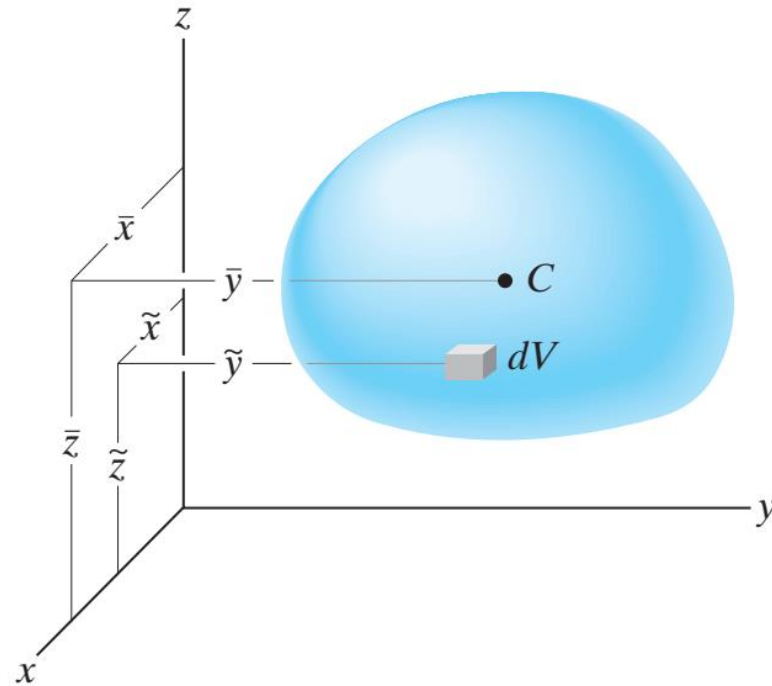
$$\bar{z} = \frac{\int \tilde{z} \, dm}{\int dm}$$

$$dm = \rho \, dv$$

$$\frac{\int \tilde{x} \rho \, dv}{\int \rho \, dv}$$

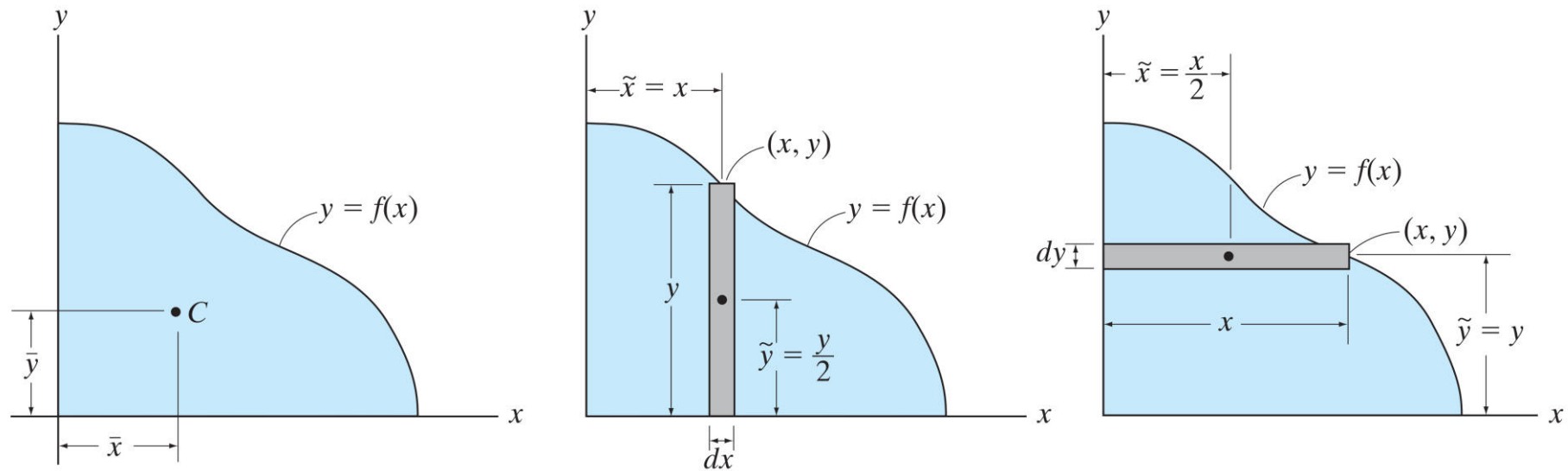


Centroid of a Volume



$$\bar{x} = \frac{\int_V \tilde{x} dV}{\int_V dV} \quad \bar{y} = \frac{\int_V \tilde{y} dV}{\int_V dV} \quad \bar{z} = \frac{\int_V \tilde{z} dV}{\int_V dV}$$

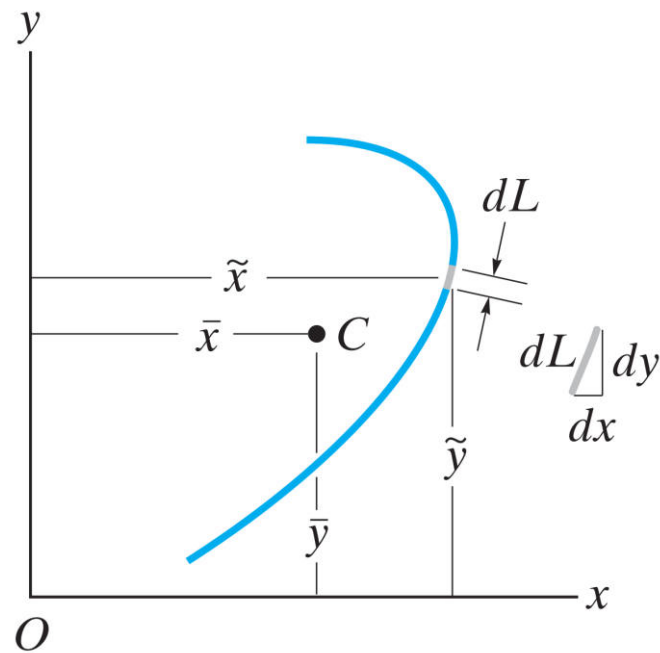
Centroid of an Area



$$\bar{x} = \frac{\int_A \tilde{x} dA}{\int_A dA}$$

$$\bar{y} = \frac{\int_A \tilde{y} dA}{\int_A dA}$$

Centroid of a Line



$$\bar{x} = \frac{\int_L \tilde{x} dL}{\int_L dL}$$

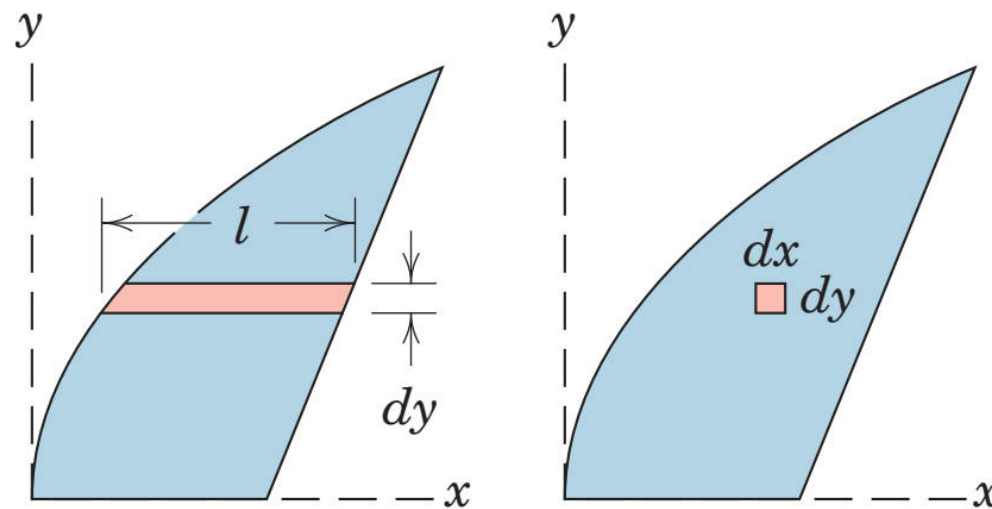
$$\bar{y} = \frac{\int_L \tilde{y} dL}{\int_L dL}$$

Concluding Notes

- The centroid represents the geometric center of a body. This point coincides with the center of mass or the center of gravity only if the material composing the body is uniform or homogeneous.
- Formulas used to locate the center of gravity, or the centroid simply represent a balance between the sum of moments of all the parts of the system and the moment of the “resultant” for the system.
- In some cases, the centroid is located at a point that is not on the object, as in the case of a ring, where the centroid is at its center. Also, this point will lie on any axis of symmetry for the body.

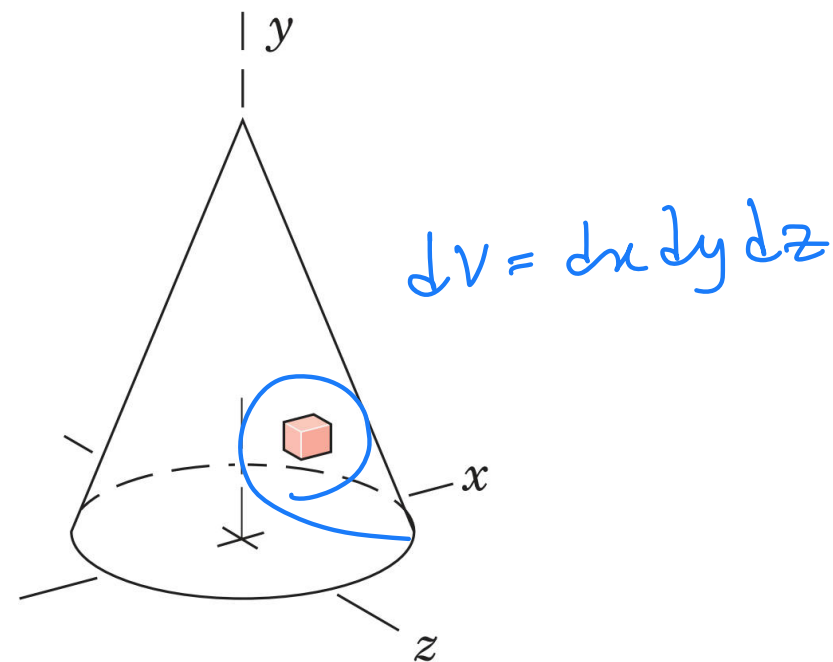
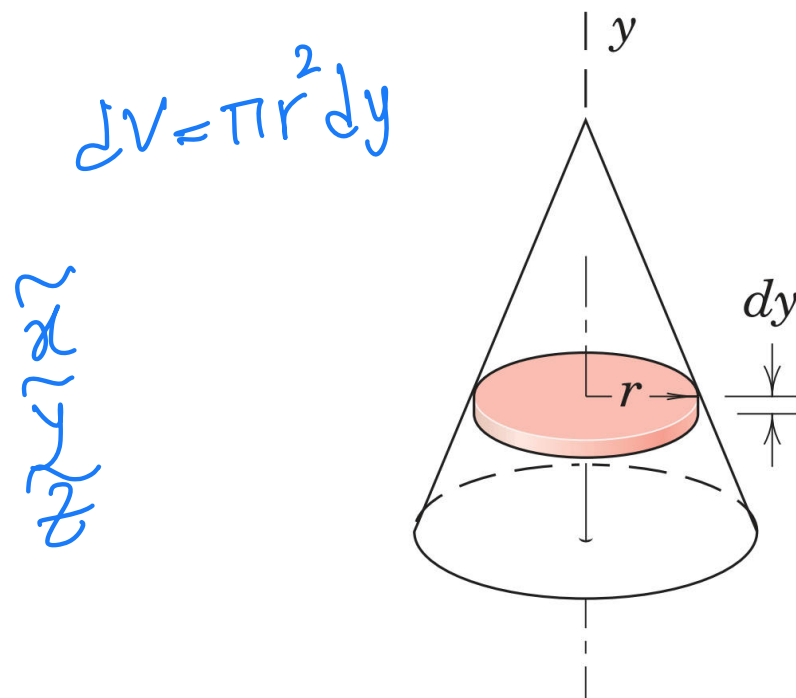
Choice of Element for Integration

(1) Order of Element. Whenever possible, a first-order differential element should be selected in preference to a higher-order element so that only one integration will be required to cover the entire figure.



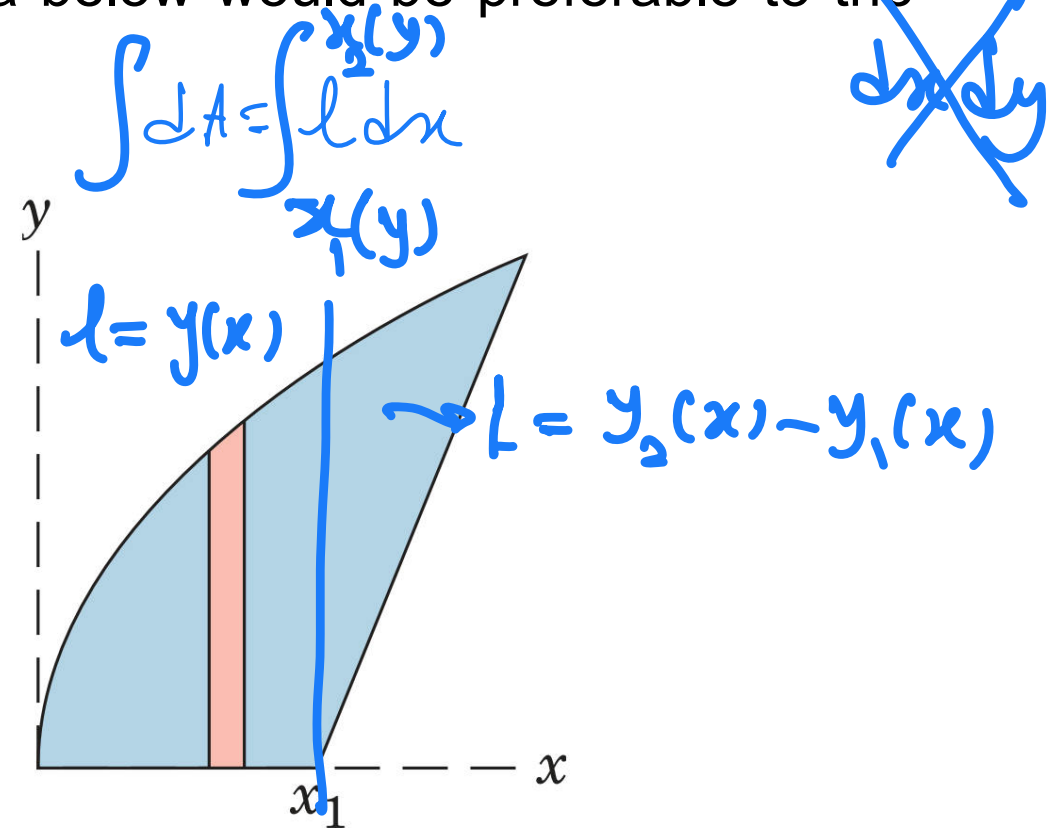
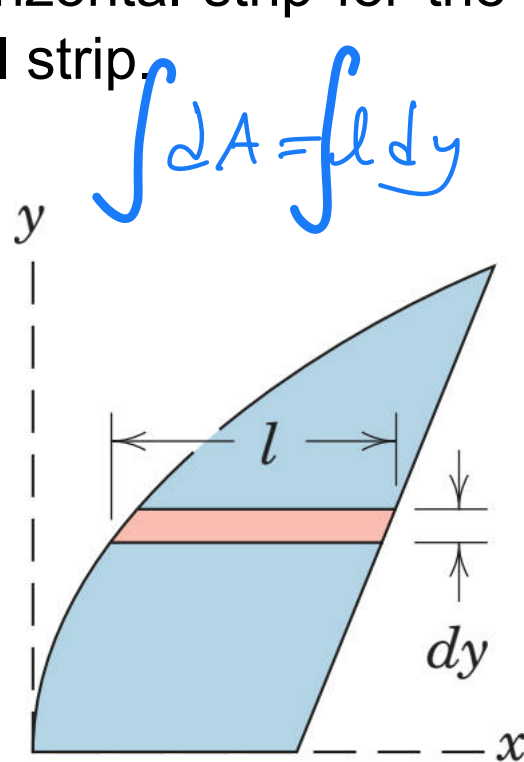
Choice of Element for Integration

- (1) Order of Element.** Whenever possible, a first-order differential element should be selected in preference to a higher-order element so that only one integration will be required to cover the entire figure.



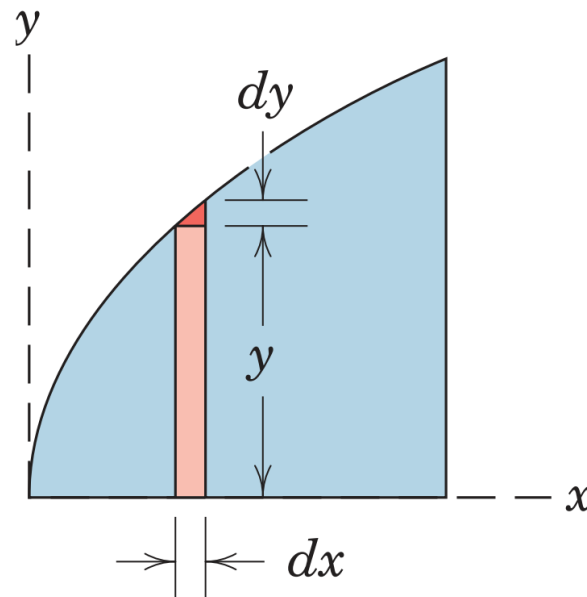
Choice of Element for Integration

(2) Continuity. Whenever possible, we choose an element which can be integrated in one continuous operation to cover the figure. Thus, the horizontal strip for the area below would be preferable to the vertical strip.



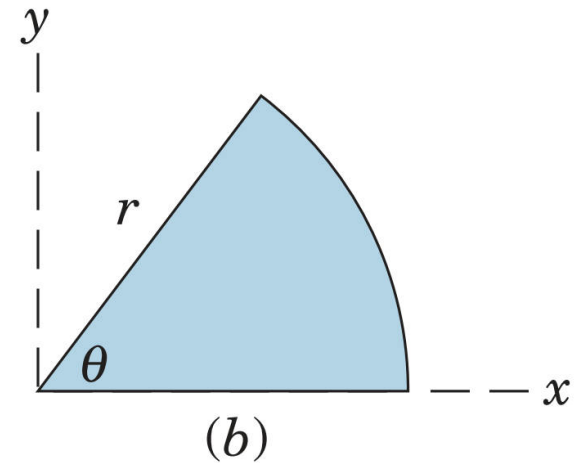
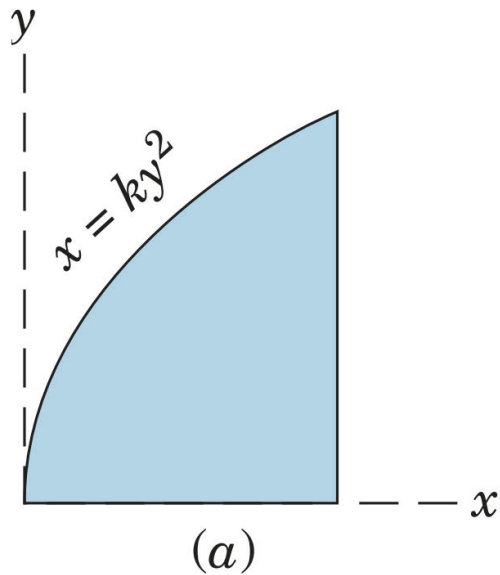
Choice of Element for Integration

(3) Discarding Higher-Order Terms. Higher-order terms may always be dropped compared with lower-order terms. Thus, the vertical strip of area under the curve below is given by the first-order term $dA = y \, dx$, and the second-order triangular area $dx \, dy/2$ is discarded. In the limit, of course, there is no error.



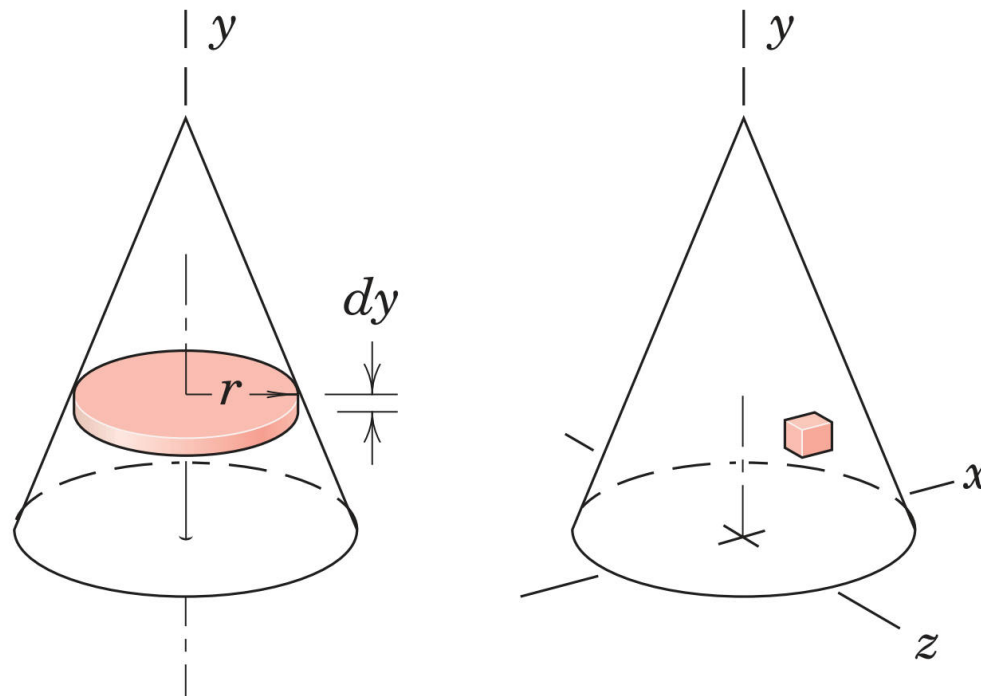
Choice of Element for Integration

(4) Choice of Coordinates. Generally, we choose the coordinate system which best matches the boundaries of the figure.



Choice of Element for Integration

(5) Centroidal Coordinate of Element. When a first- or second- order differential element is chosen, it is essential to use the *coordinate of the centroid of the element* for the moment arm in expressing the moment of the differential element.



Example: Locate the centroid of the rod bent into the shape of a parabolic arc as shown below.

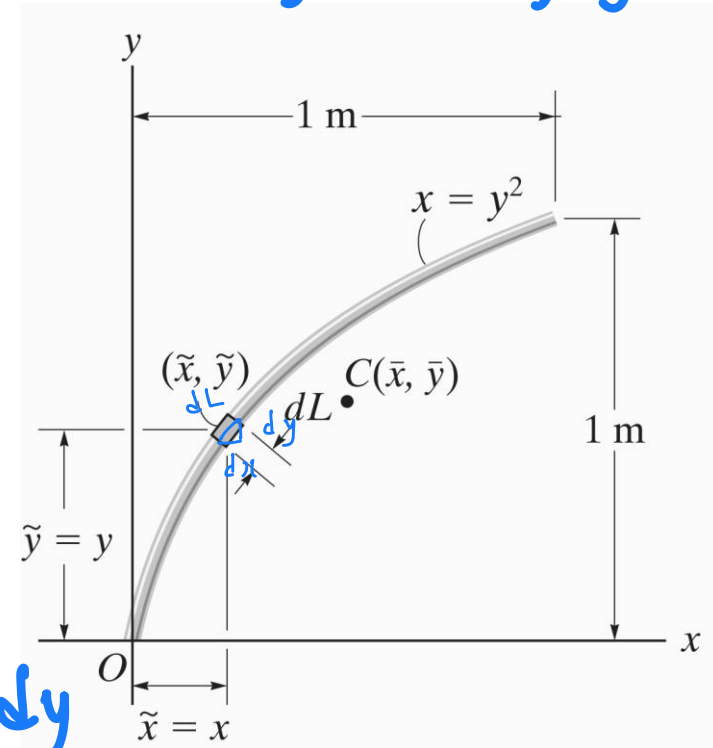
$$x = y^2 \rightarrow 2y dy = dx$$

$$\bar{x} = \frac{\int \tilde{x} dL}{\int dL} ; dL^2 = dx^2 + dy^2$$

$$L = \int dL = \int \sqrt{dx^2 + dy^2}$$

$$= \int dy \sqrt{\left(\frac{dx}{dy}\right)^2 + 1} = \int \sqrt{(2y)^2 + 1} dy$$

$$\bar{x} = \frac{1}{L} \int x \sqrt{dx^2 + dy^2} = \frac{1}{L} \int_0^1 y^2 \sqrt{4y^2 + 1} dy$$



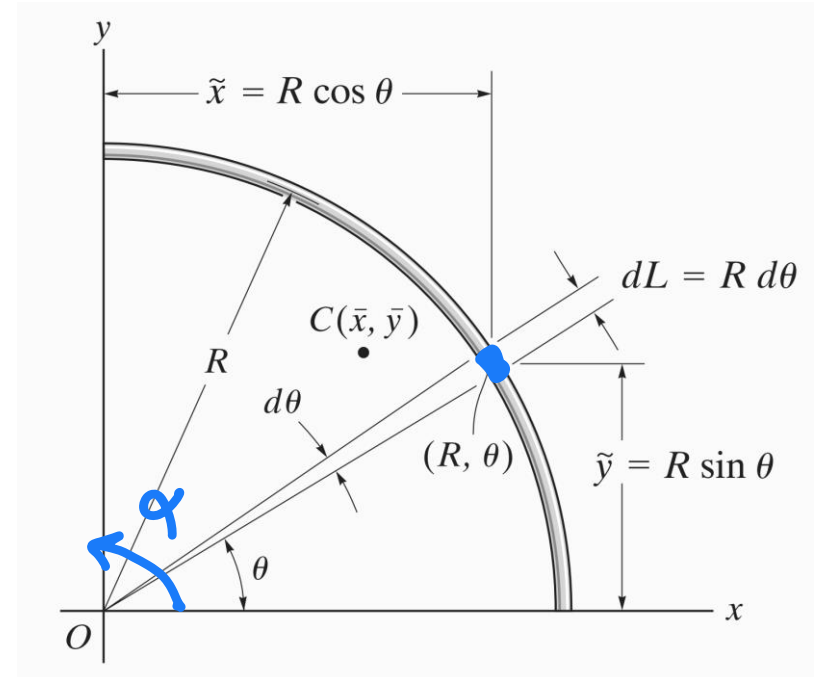
Example: Locate the centroid of the circular wire segment shown below.

$$\bar{x} = \frac{\int \tilde{x} dL}{\int dL} = \frac{\int R \cos \theta (R d\theta)}{\int R d\theta}$$

$$L = \frac{1}{4} (2\pi R) = \frac{\pi}{2} R \rightarrow \frac{\alpha}{2\pi} (2\pi R)$$

$$\bar{x} = \frac{2}{\pi R} \int_0^{\frac{\pi}{2}} R^2 \cos \theta d\theta =$$

$$\frac{2}{\pi R} \left(R^2 \sin \frac{\pi}{2} \right) = \frac{2R}{\pi} \rightarrow$$



$$\frac{R \sin \alpha}{\alpha}$$

Example: Determine the distance y measured from the x axis to the centroid of the area of the triangle shown below.

$$\frac{x}{h-y} = \frac{b}{h}$$

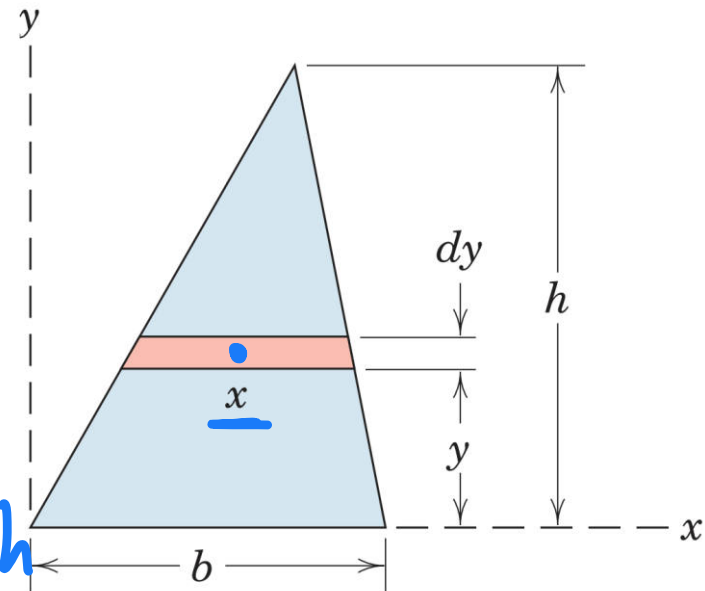
$$dA = x dy$$

$$\tilde{y} = y$$

$$\bar{y} = \frac{\int \tilde{y} dA}{\int dA}$$

$$\int dA = \int x dy = \int_0^h \frac{b}{h} (h-y) dy = \frac{1}{2} bh$$

$$\int y dA = \int y x dy = \int_0^h y \frac{b}{h} (h-y) dy = \frac{bh^2}{3}$$



Example: Locate the centroid for the area of a circular sector shown below. (Use two different **first**-order elements)

$$\bar{x} = \frac{\int \tilde{x} dA}{\int dA} \quad \int dA = A = \frac{2\alpha}{2\pi} (\pi r^2) = \alpha r^2$$

$$\tilde{x} = \left(\frac{2}{3}r\right) \cos \theta$$

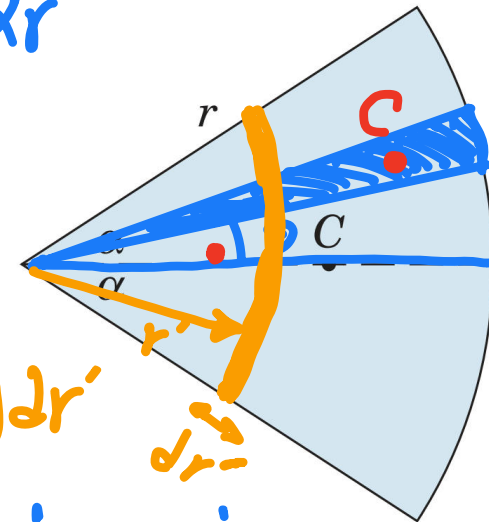
$$dA = \frac{1}{2}r (r d\theta)$$

$$\bar{x} = \frac{\int_{-\alpha}^{\alpha} \frac{1}{2} r^2 \left(\frac{2}{3}r\right) \cos \theta d\theta}{\alpha r^2}$$



$$dA = (2\alpha r') dr'$$

$$\tilde{x} = \frac{r' \sin \alpha}{\alpha}$$

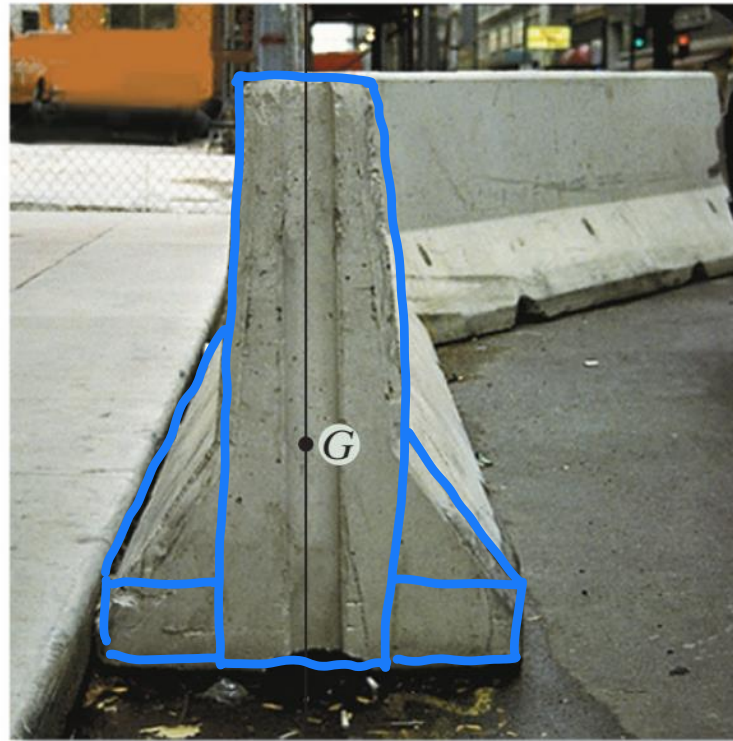


Composite Bodies



A stress analysis of this angle requires that the centroid of its cross-sectional area be located.

Composite Bodies



In order to determine the force required to tip over this concrete barrier, it is first necessary to determine the location of its center of gravity G . This point will lie on the vertical axis of symmetry.

Composite Bodies

The moment principle

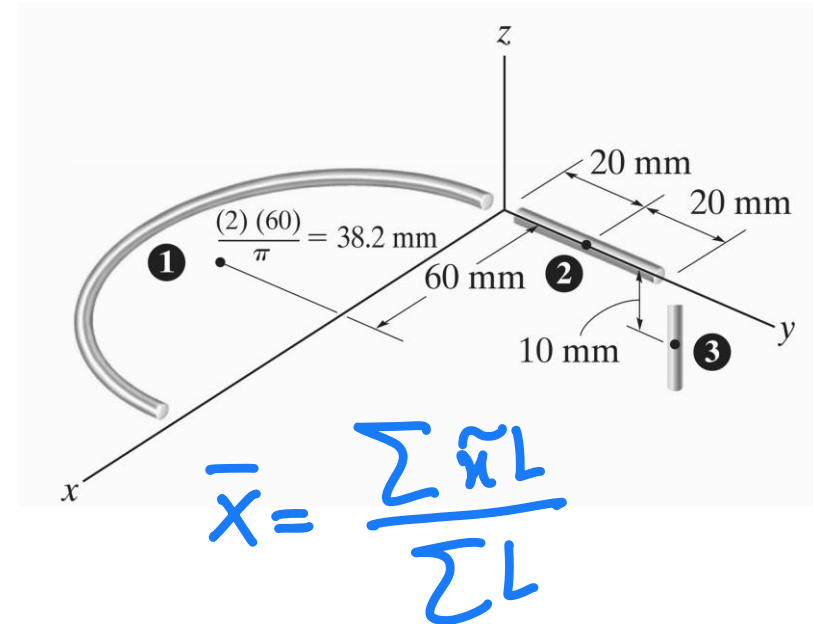
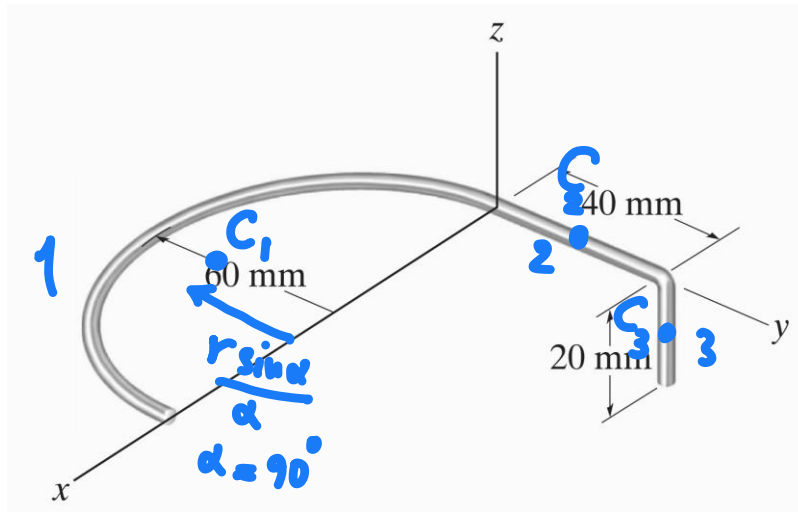
$$\bar{x} = \frac{\sum \tilde{x} W}{\sum W} \quad \bar{y} = \frac{\sum \tilde{y} W}{\sum W} \quad \bar{z} = \frac{\sum \tilde{z} W}{\sum W}$$

$\bar{x}, \bar{y}, \bar{z}$ represent the coordinates of the center of gravity G of the composite body.

$\tilde{x}, \tilde{y}, \tilde{z}$ represent the coordinates of the center of gravity of each composite part of the body.

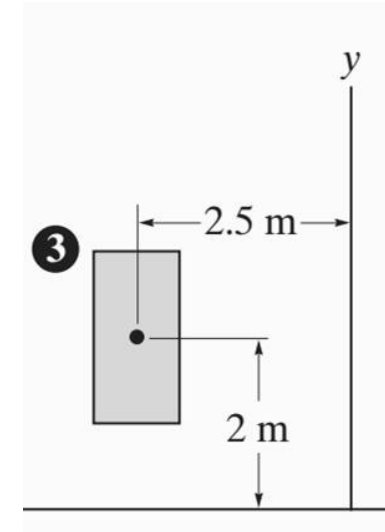
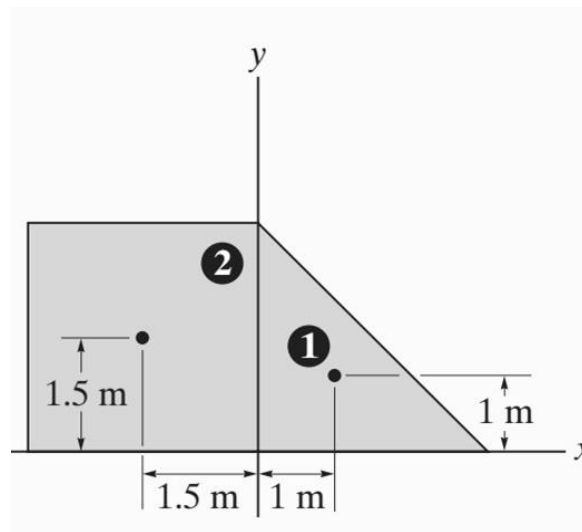
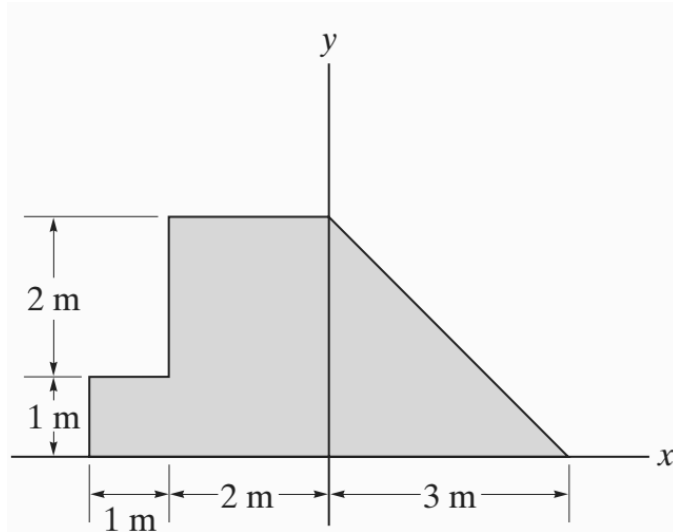
$\sum W$ is the sum of the weights of all the composite parts of the body, or simply the total weight of the body.

Example: Locate the centroid of the wire shown below.



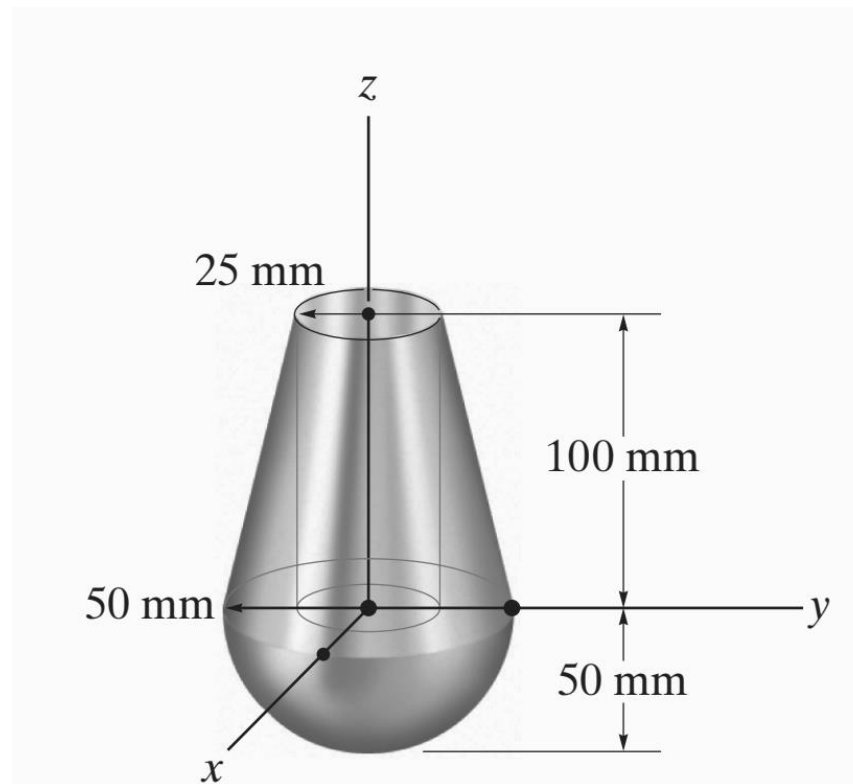
Segment	L (mm)	$\tilde{x}$ (mm)	$\tilde{y}$ (mm)	$\tilde{z}$ (mm)	$\tilde{x}L$ (mm ²)	$\tilde{y}L$ (mm ²)	$\tilde{z}L$ (mm ²)
1	$\pi(60) = 188.5$	60	-38.2	0	11 310	-7200	0
2	40	0	20	0	0	800	0
3	20	0	40	-10	0	800	-200
	$\Sigma L = 248.5$				$\Sigma \tilde{x}L = 11\,310$	$\Sigma \tilde{y}L = -5600$	$\Sigma \tilde{z}L = -200$

Example: Locate the centroid of the plate area shown below.

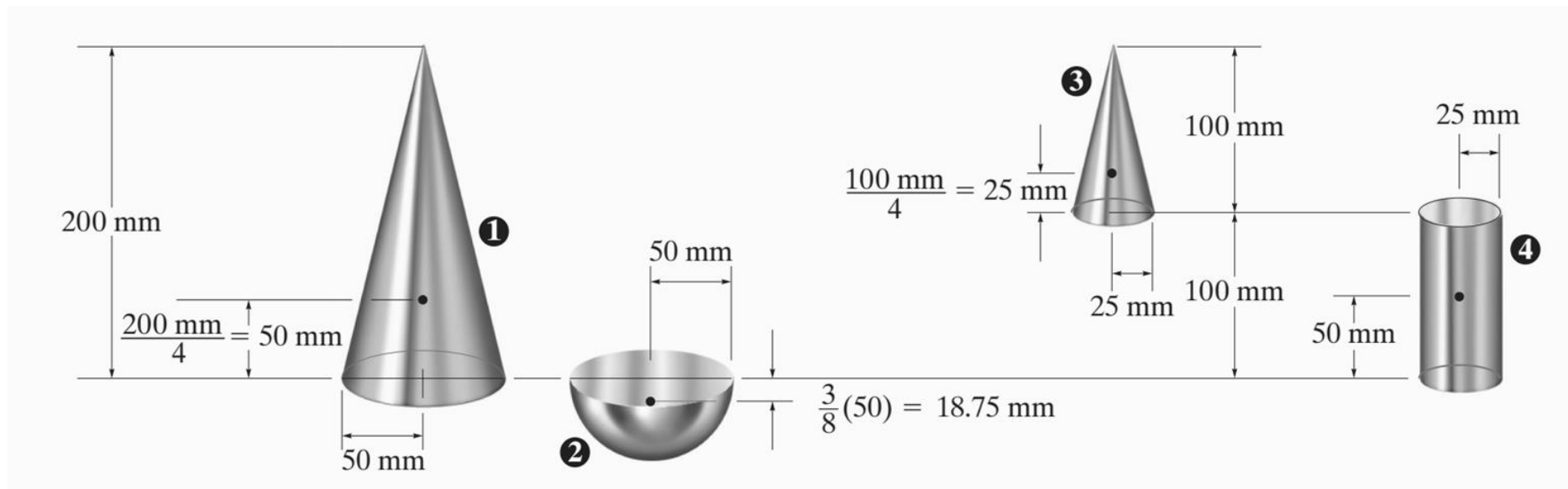


Segment	$A \text{ (m}^2\text{)}$	$\tilde{x} \text{ (m)}$	$\tilde{y} \text{ (m)}$	$\tilde{x}A \text{ (m}^3\text{)}$	$\tilde{y}A \text{ (m}^3\text{)}$
1	$\frac{1}{2}(3)(3) = 4.5$	1	1	4.5	4.5
2	$(3)(3) = 9$	-1.5	1.5	-13.5	13.5
3	$-(2)(1) = -2$	-2.5	2	5	-4
	$\Sigma A = 11.5$			$\Sigma \tilde{x}A = -4$	$\Sigma \tilde{y}A = 14$

Example: Locate the center of mass of the assembly shown below. The conical frustum has a density of $\rho_c = 8 \text{ Mg/m}^3$, and the hemisphere has a density of $\rho_h = 4 \text{ Mg/m}^3$. There is a 25-mm-radius cylindrical hole in the center of the frustum.



Example: Locate the center of mass of the assembly shown below. The conical frustum has a density of $\rho_c = 8 \text{ Mg/m}^3$, and the hemisphere has a density of $\rho_h = 4 \text{ Mg/m}^3$. There is a 25-mm-radius cylindrical hole in the center of the frustum.

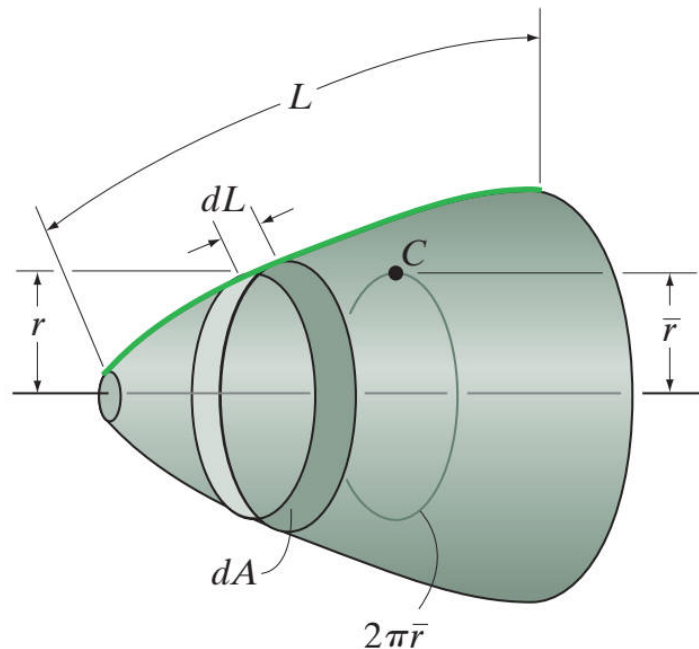


Theorems of Pappus

The two *theorems of Pappus* are used to find the surface area and volume of any body of **revolution**.

Surface Area

If we revolve a *plane curve* about an axis that does not intersect the curve, we will generate a *surface area of revolution*.



$$A = 2\pi \int r \, dL$$

Theorems of Pappus

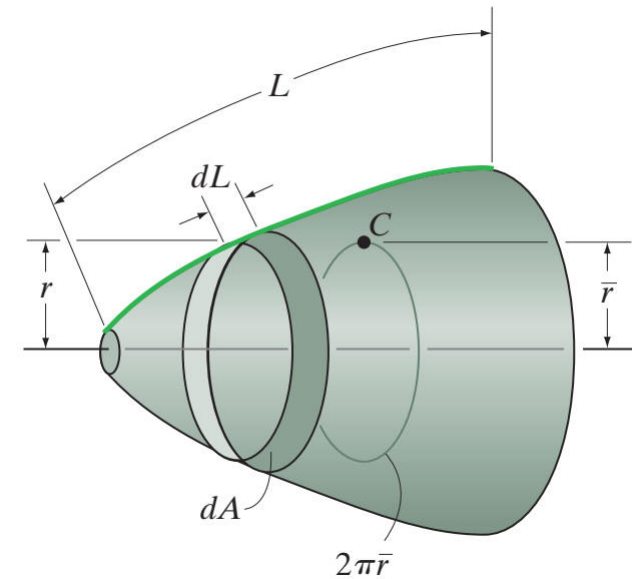
$$A = \theta \bar{r} L$$

A = surface area of revolution

θ = angle of revolution measured in radians, $\theta \leq 2\pi$

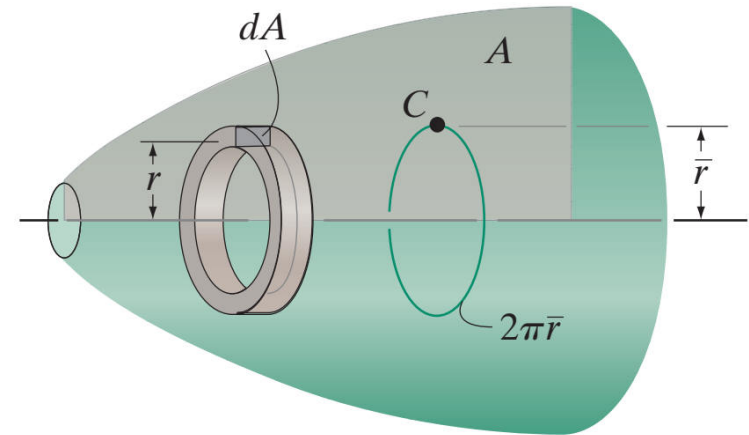
$\bar{r}$ = perpendicular distance from the axis of revolution to the centroid of the generating curve

L = length of the generating curve



Theorems of Pappus

Volume



$$V = \theta \bar{r} A$$

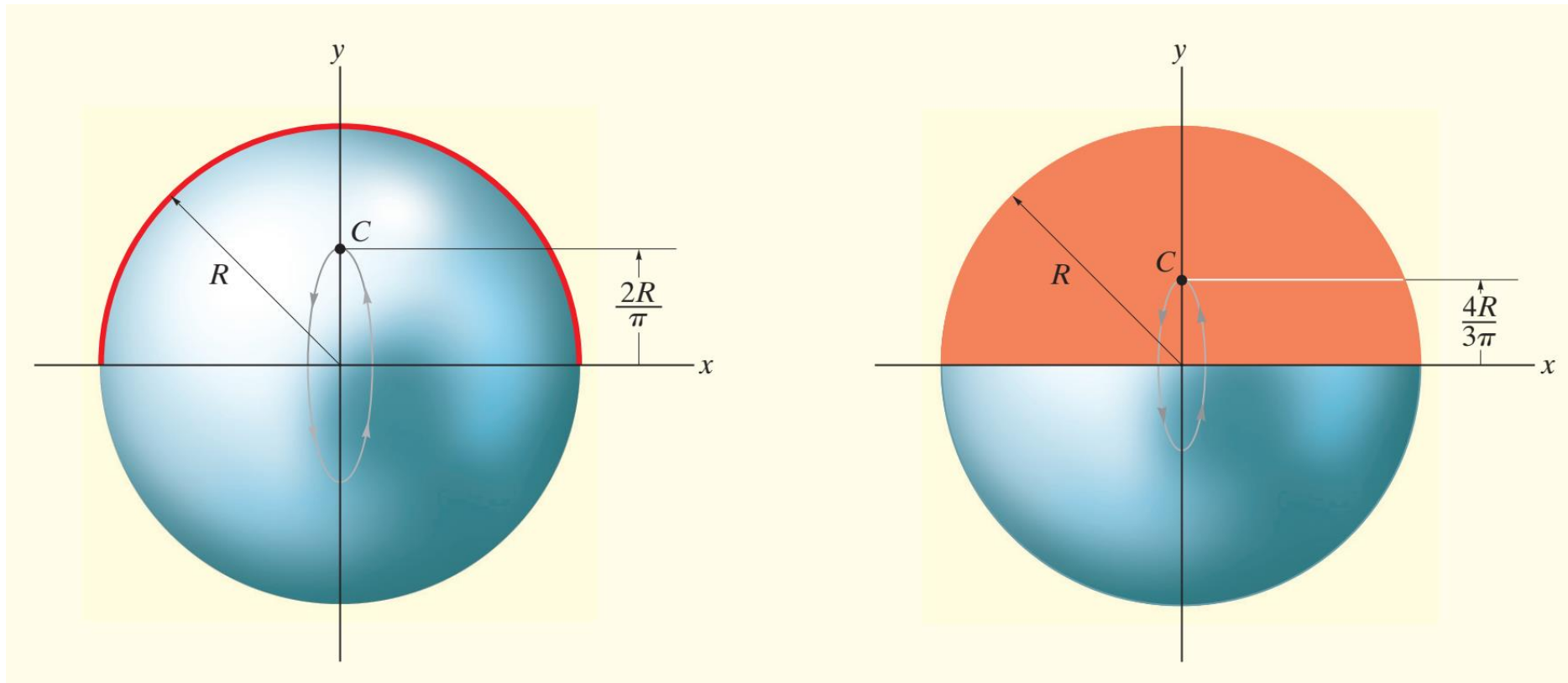
A = surface area of revolution

θ = angle of revolution measured in radians, $\theta \leq 2\pi$

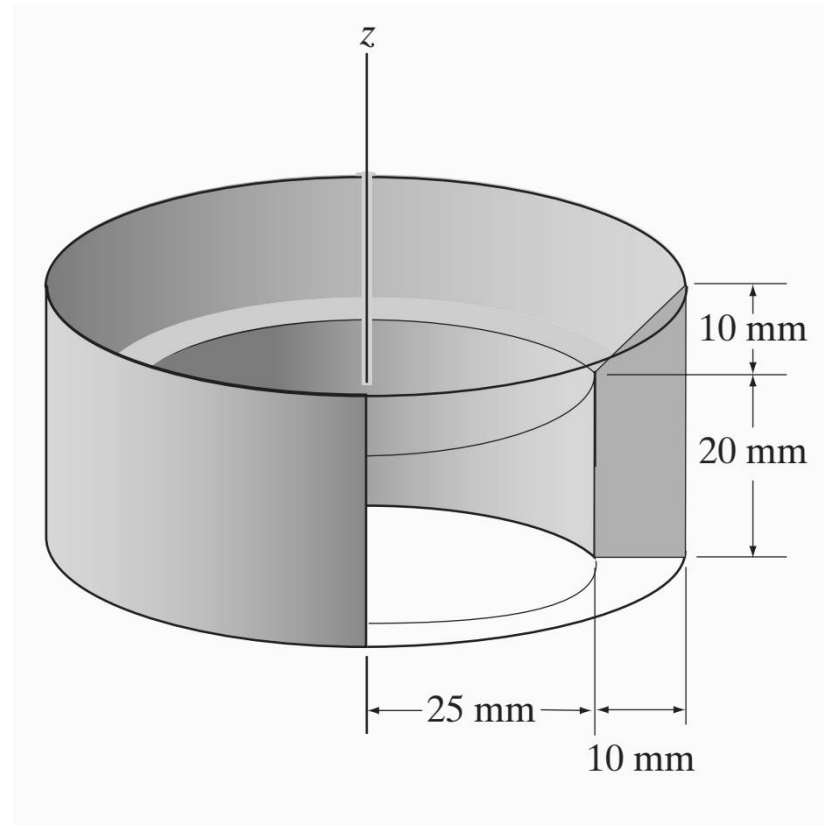
$\bar{r}$ = perpendicular distance from the axis of revolution to the centroid of the generating curve

L = length of the generating curve

Example: surface and volume of a sphere.

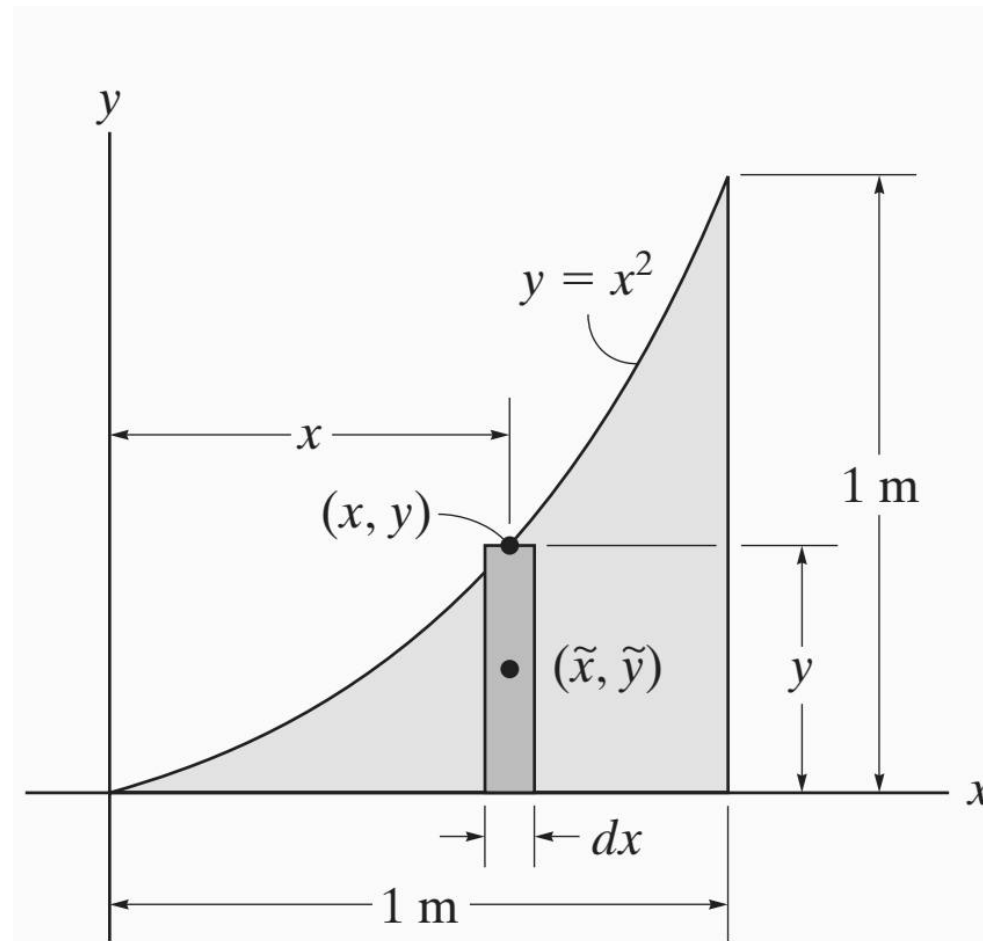


Example: Determine the surface area and volume of the full solid shown below.



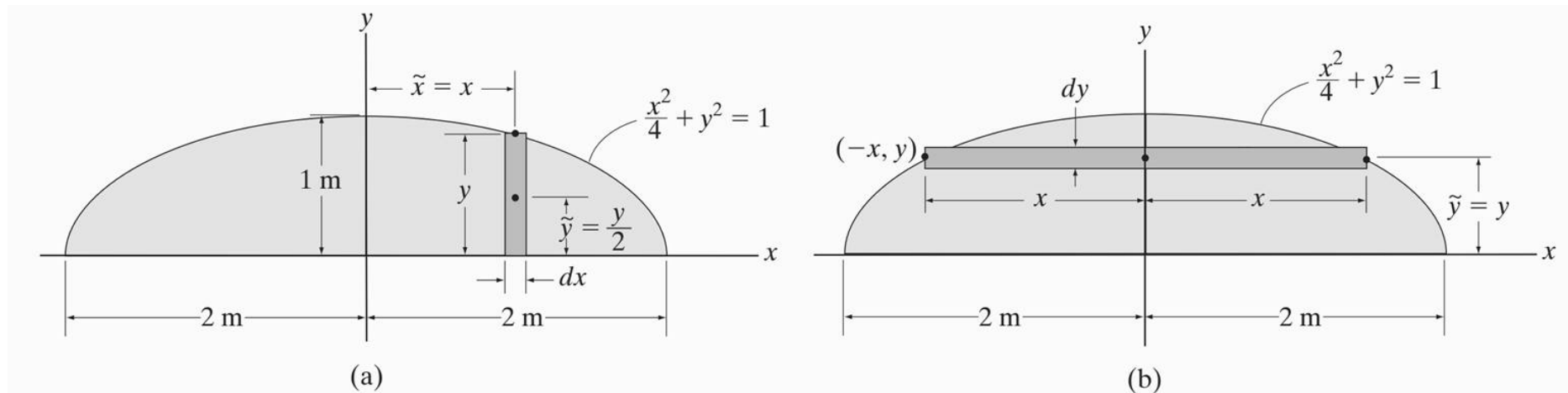
Practice at Home

1. Locate the centroid of the area shown below.



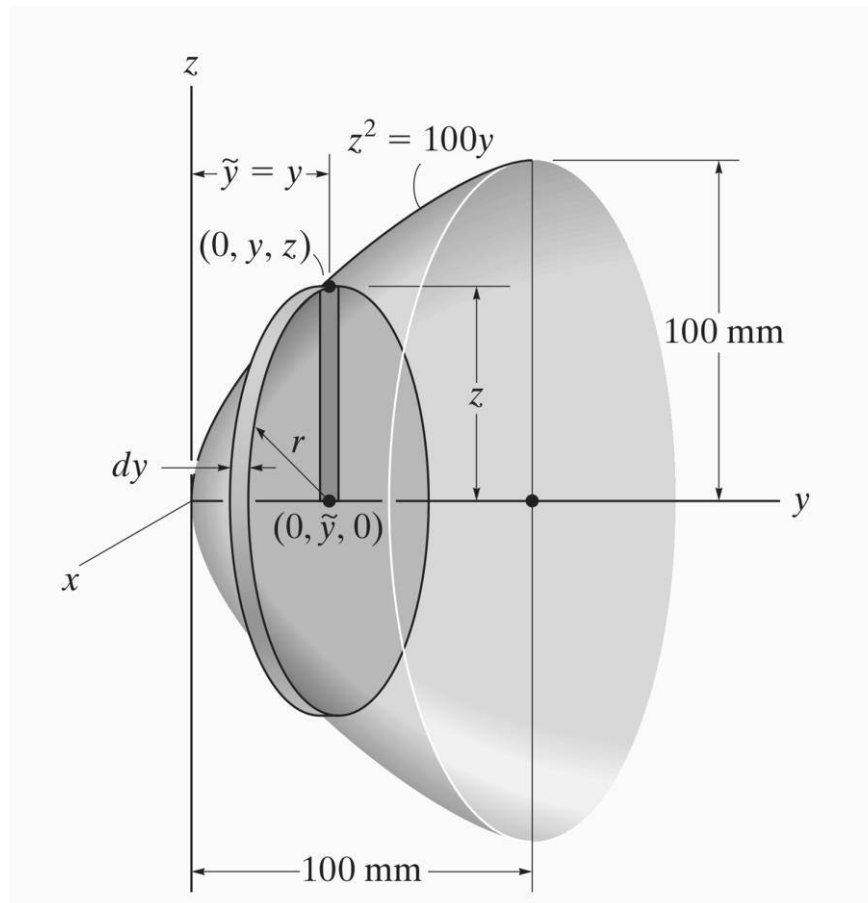
Practice at Home

2. Locate the centroid of the semi-elliptical area shown below based on both elements shown.



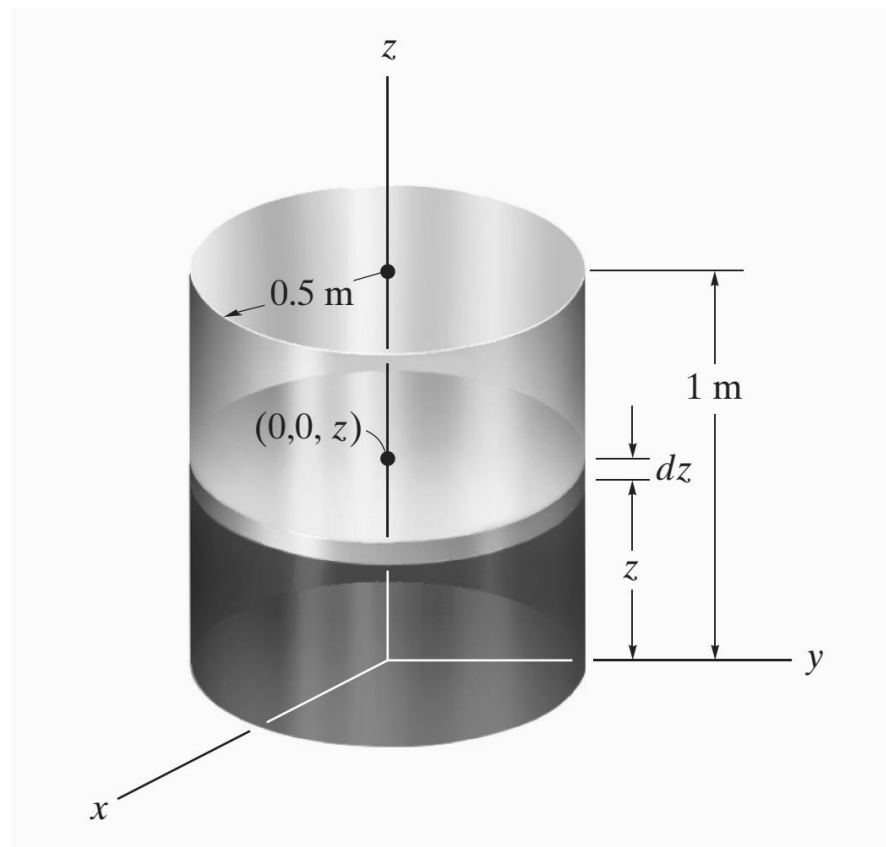
Practice at Home

3. Locate the y centroid for the paraboloid of revolution, shown below.



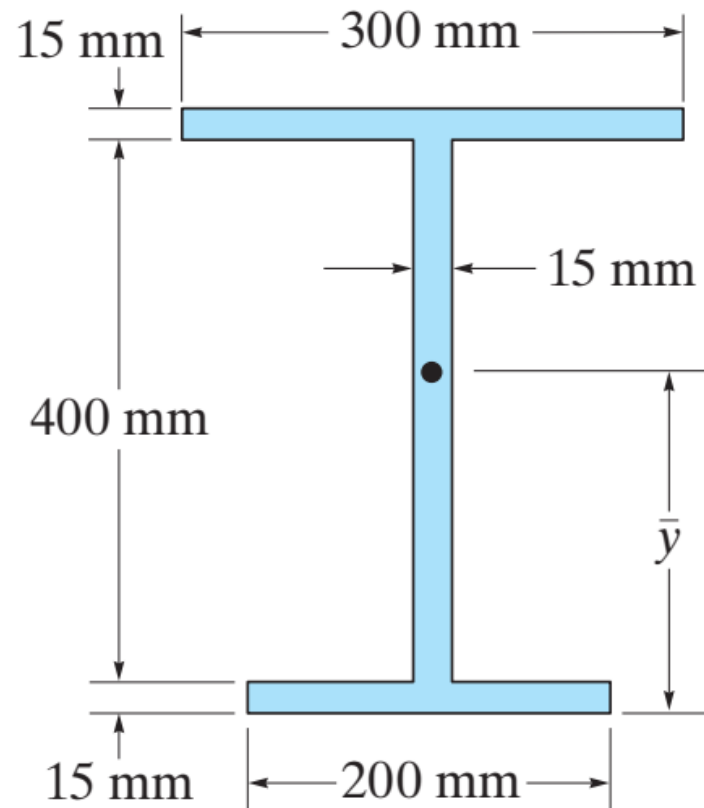
Practice at Home

4. Determine the location of the center of mass of the cylinder shown below if its density varies directly with the distance from its base, i.e., $\rho = 200z \text{ kg/m}^3$.



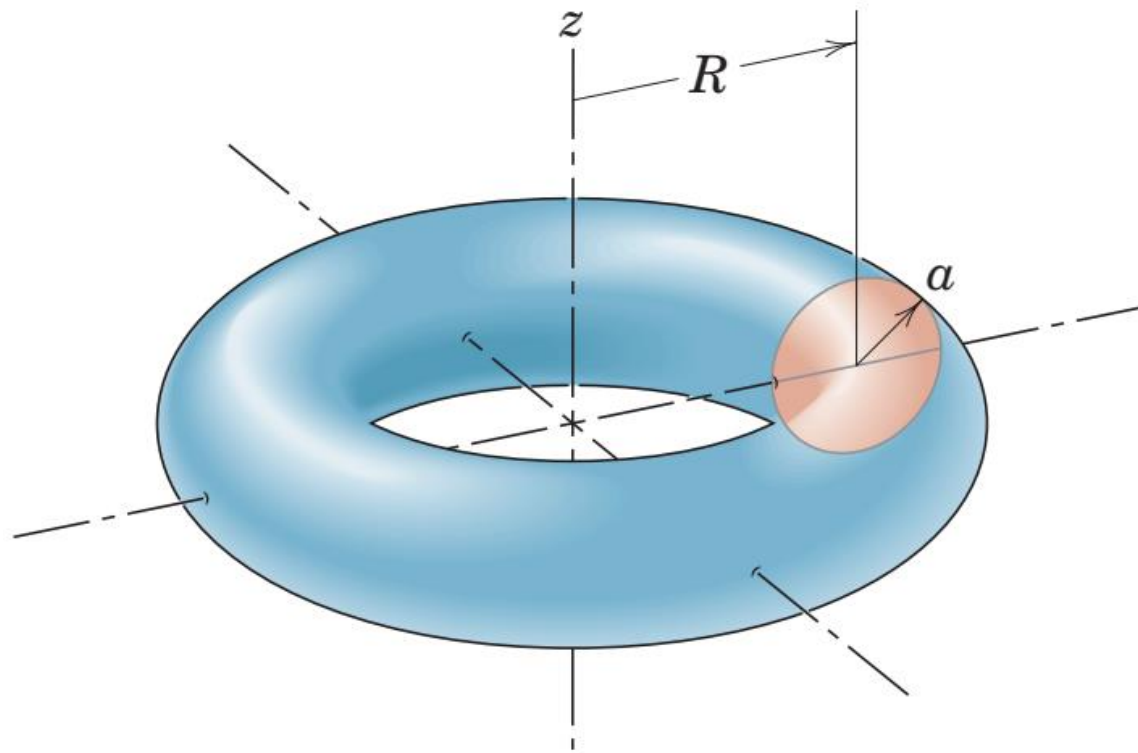
Practice at Home

5. Determine the location $\bar{y}$ of the centroid for the beam's cross-sectional area.



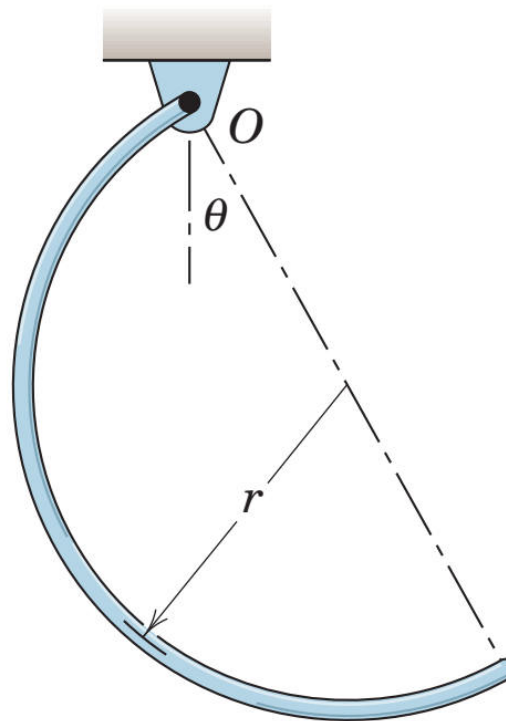
Practice at Home

6. Determine the volume V and surface area A of the complete torus of circular cross section.



Practice at Home

7. A uniform semicircular rod of radius r is supported in a bearing at its upper end and is free to swing in the vertical plane. Calculate the angle θ made by the diameter with the vertical for the equilibrium position.



Practice at Home

8. In order to provide sufficient support for the stone masonry arch designed as shown, it is necessary to know its total weight W . Determine W and the center of gravity. The density of stone masonry is 2.4 Mg/m^3 .

